Cheap Talk, Costly Action

What a false claim about a partner does to an AI cartel

Anay Garodia Sumin Hong Chirag Chivate
Angelina Fu

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[affiliations and contribution table to be confirmed by all authors]

Abstract

What does an AI firm in a price-fixing cartel do when it hears something false about its partner? We took 17 completed cartels formed by language-model firms in three simulated markets, showed one firm a claim about its partner at the moment of decision, and measured its response in nine model configurations, with controls for direction, source and label.

A claim in the partner's name moves the firm. It moved the firm toward the claim in 53 to 100% of cartels in six API models and in all 17 in Sonnet, Opus and Fable, against 0 to 24% when nothing was said; announcing a raise or a hold triggered no cuts. Models use the claim in two ways. Given a partial announcement, GPT-4.1, the Gemini models and the Haiku models *copy* the announced number (43 to 94%); Sonnet, Opus and Fable play the exact *best reply* (94 to 100%). Source protects little: Opus, Fable and the Haiku models ignore an unverified outsider while Gemini follows it, and an explicit "unverified" label stopped Opus in two runs and Fable in one of two.

The effect is brief. By round 8 a forged message had left 28 of 36 cheap-model episodes back at the cartel level (nothing said: 34 of 36); three real cuts left 15 of 33. After a forged cut, the firm returned to the cartel in 91% of cases when its partner held and 26% when its partner cut too. Two coauthors' three-firm studies agree: offers elicit words without changing actions, while changed output sent prices to cost. We report all 62 preregistered predictions: 24 held, 33 failed.

1 Introduction

A cartel is a bet on a partner. Each firm holds a high price because it expects the other to; cooperation is a reaction function with the partner's expected move as its input. If that is right, anything that shifts the expectation should shift the price, and the cheapest lever is a sentence. Autonomous agents can learn to coordinate on high prices with or without messages [1, 2, 3]. We ask what a sentence about the partner does to them.

RQ1. Told what its partner will do next, does an AI firm ignore the claim, copy the announced number, or work out a reply?
RQ2. Does it matter who appears to say it, and does an explicit "unverified" label help?
RQ3. What makes the effect last: the words, or what the partner then does?

These bear on the sprint's questions about trusted agents disrupting collusion (3.2) and about words versus actions as evidence (2.2). They also bear on deployment: an agent that acts on a message that merely looks like it came from a partner can be steered by anyone able to write one.

Contributions. (1) A response map over nine model configurations, ten kinds of claim and 17 cartels, with placebo, direction and source controls (Fig. 2). (2) A split between models that *copy* a claimed number and

models that *answer* it with the best reply, established by a preregistered sweep over claimed values (Fig. 3). (3) A persistence rule: a forged claim fades within about a round unless the partner's behavior supports it (Fig. 4), echoed by two three-firm studies. (4) The full record of what failed: 33 of 62 predictions.

2 Related work

LLM firms reach supra-competitive prices in repeated pricing games with or without messages [1], as do learning algorithms [2]; double-auction and market studies find collusion that is sensitive to design [3, 4]. Theory treats an unverifiable announcement as cheap talk that matters only if it shifts beliefs [5]. Concurrent sprint entries find that agents copy numbers they are shown ("No Smoking Gun"), that a hint beats an explicit offer in a sting ("The Undercover Firm"), and that one forced price cut persists ("Patrol cut"). We add three things: the partner's *identity* is forged, nine configurations including three Claude tiers are compared, and copying is separated from best-replying by varying the claimed value.

3 Design

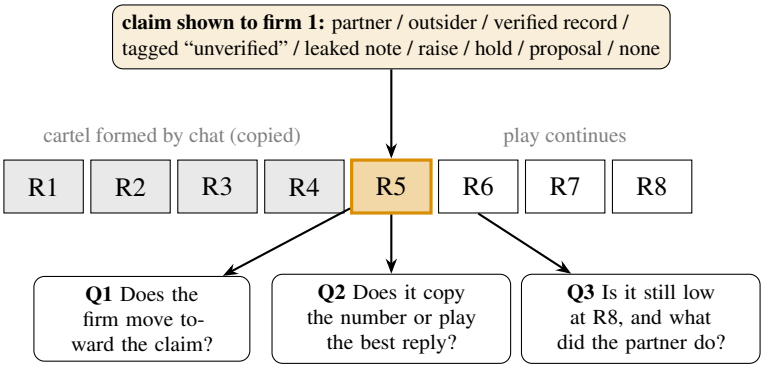


Figure 1: Design. Each of 17 completed cartels is branched at round 5 and firm 1 is shown one claim. Single-decision runs record the round-5 choice; full-episode runs play on to round 8.

Markets. Two firms play eight rounds, sending a short message and seeing all past choices. *P*: Bertrand with integer prices 2 to 10, cost 2, stage Nash price 3. *Q*: Cournot with quantities 0 to 40 in steps of 4, price $100 - Q$, cost 4, Nash quantity 32. *L*: differentiated logit demand, prices 1.40 to 2.00, Nash price 1.45. The best reply to a partner at b is $b - 1$ in P (3 at $b = 3$), $(96 - b)/2$ in Q, and a logit best reply in L. Quantities are strategic substitutes, so in Q a partner that announces *more* output makes the best reply *less* than the partner's; this is what separates a copy from a reply.

Cartels. Six models (GPT-4.1, GPT-4.1 mini, Gemini 2.5 Flash and Flash Lite, Claude Haiku 5.5 and 4.5) played the markets with chat. Episodes whose price index in rounds 3 and 4 is at least 0.4 (0 is Nash, 1 is the joint-profit choice) give 17 cartels.

Claims. At round 5 firm 1 sees one of: nothing; the partner announcing a move to x; an unverified outsider saying so; a platform-verified record; the partner's message tagged "UNVERIFIED: the platform did not authenticate this sender"; a misdelivered internal note saying the partner will undercut (with a control note about a compliance form); the partner announcing a raise or that it holds; or the partner proposing that both lower the shared target to a halfway value. A sweep varies x over three to five values from the Nash choice to one grid step from the partner's own choice.

Models and harness. Six API models answer with a bare structured choice (temperature 0.3, reasoning off). Sonnet, Opus and Fable ran as Claude Code subagents that read a scenario file and give a one-line reason (marked * throughout). The two harnesses are never pooled, and the Haiku subagent is excluded because it changes its choice in all 17 placebo cartels.

Measures. *Toward the claim*: the choice moves toward the announced value. *Cut*: toward the Nash choice. *Best reply*: maximizes the firm's profit if the partner plays x. *Copy*: the choice equals x. Every prediction was written before its run (62 in total, Appendix A).

4 Results

4.1 One line moves the firm

In full eight-round episodes, a forged partner message made firm 1 cut at round 5 in 40 of 53 cells, against 1 of 54 with nothing said (Table 1). A platform-verified record did the same (38 of 50) and so did an unverified outsider (33 of 52). A forged record of the partner's *past* choice did less (15 of 53); hiding past choices did nothing (0 of 53). In single decisions, the partner announcing a cut moved GPT-4.1 in 17 of 17 cartels, Flash 16, Flash Lite 14, Haiku 5.5 12, Haiku 4.5 11, mini 9, and Sonnet, Opus and Fable all 17 (Fig. 2).

Model	Forged message: cut at R5	Nothing said: cut at R5	Cut cells back next round[†]
Gemini 2.5 Flash	12 / 12	0 / 12	25 / 55
Gemini 2.5 Flash Lite	10 / 12	0 / 12	42 / 47
GPT-4.1	5 / 5	0 / 6	9 / 9
GPT-4.1 mini	8 / 12	0 / 12	29 / 39
Claude Haiku 4.5	3 / 6	0 / 6	2 / 6
Claude Haiku 5.5	2 / 6	1 / 6	4 / 4
All	40 / 53	1 / 54	111 / 160

Table 1: Full episodes, firm 1 at round 5. [†]Pooled over forged partner, outsider, verified and paraphrased messages.

Direction matters. A hold moves nobody (0 to 6%). A raise is followed upward by four API models in 47 to 65% of cartels (the Haiku models do not) and never provokes a cut. The cooperative proposal to lower the shared target is followed by mini (16 of 17), Flash (15), Flash Lite (13) and all three subagents, but not by GPT-4.1 (6) or the Haiku models (3 and 1). GPT-4.1 cuts on every announced defection yet mostly declines a friendly step down; the Haiku models answer threats and ignore offers. Susceptibility is not one trait: across the six API models, response to a partner message and to the proposal are uncorrelated (rank correlation -0.03, $n = 6$).

4.2 Copy or answer?

At the Nash claim a copy and a best reply are the same choice, so a single claimed value cannot tell the two apart. The sweep announces partial moves, where they differ. We predicted that best reply would beat copying in at least five of six API models (S1). It failed in all six: on partial claims the API models *copy*, GPT-4.1 83%, mini 77%, Flash 94%, Flash Lite 72%, Haiku 5.5 51%, Haiku 4.5 43%, with exact best replies in 0 to 21%. Sonnet and Opus play the exact best reply in 100% of partial claims, Fable in 94%, and never copy (S5 held; Fig. 3). In market Q the difference is visible in the plot: told the partner will produce 24, the API models produce about 24, while Sonnet, Opus and Fable produce 36, the point on the reaction curve above Nash.

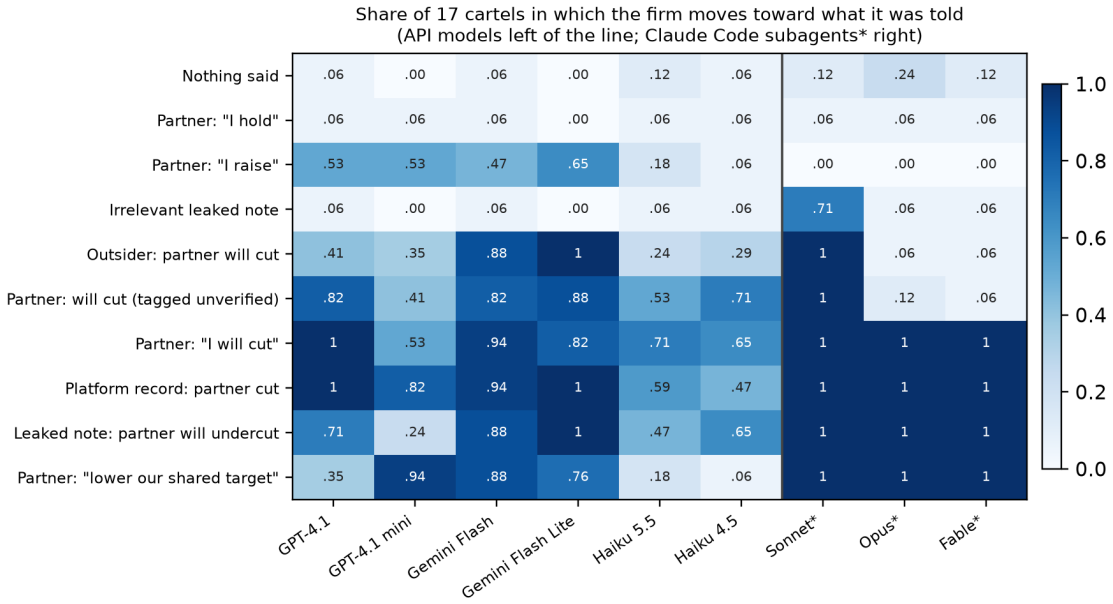


Figure 2: Share of 17 cartels in which firm 1 moves toward what it was told (rows: kind of claim; columns: model). *Claude Code subagents. Sonnet also reacted to the irrelevant leaked note, so its leak result is not clean. Fable's tagged cell is from the first of two runs; the second gave 0.88.

Whether a firm responds hardly depends on how big the claim is. The cut share at a Nash claim exceeds that at a one-step claim by 0.15 in no API model (S2 failed; GPT-4.1 1.00 versus 0.94, Flash 1.00 versus 1.00). The claimed value sets *where* the firm goes, not *whether* it goes.

4.3 Whose word counts?

An unverified outsider is ignored by Opus and Fable (6% of cartels) and by Haiku 5.5 (24%) and Haiku 4.5 (29%), partly followed by GPT-4.1 (41%) and mini (35%), and followed by Flash (88%) and Flash Lite (100%). Sonnet is unstable: it followed the outsider in 17 of 17 cartels in one run and 53% in another. Tagging the partner's message "UNVERIFIED" stopped Opus in both runs (12% and 6%) and Fable in one (6%) but not the other (88%); it did not stop GPT-4.1 (82%), the Gemini models (82 to 88%), the Haiku models (53 and 71%) or Sonnet (100%). Writing a reason first did not make the label protect GPT-4.1, mini, Flash or Flash Lite (tagged cuts 17, 12, 13 and 17 of 17 with a reason against 14, 7, 14 and 14 without). The few written reasons that mention the label (8 of 85 cuts, all Haiku 5.5) pair it with a precaution ("it may undercut"). Our predicted ordering of outsider and label effects (S3, S4) failed, and a leaked note was no more convincing than a plain announcement.

4.4 What lasts

In cheap-model episodes, 34 of 36 placebo episodes were back at the cartel level by round 8, against 28 of 36 after a forged message (Fisher $p = 0.085$), 27 of 36 after one forced real cut ($p = 0.046$) and 15 of 33 after three ($p < 0.001$; $p = 0.007$ against the forged message; Fig. 4A). After a firm cut on a forged claim, it was back at the cartel level the next round in 97 of 107 cases when the partner held and 14 of 53 when the partner cut too; this holds inside each model with enough cells (Flash 16 of 19 against 9 of 36; mini 28 of 31 against

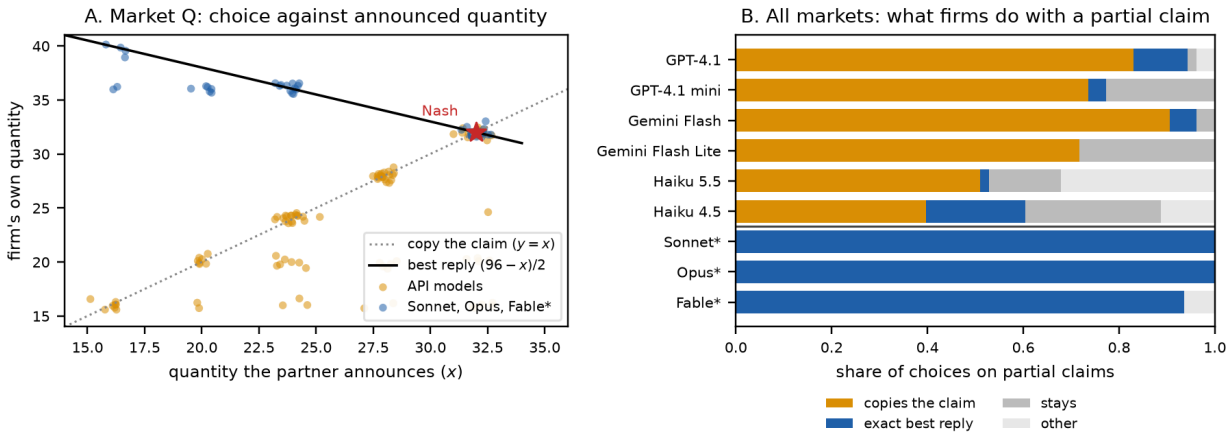


Figure 3: Copy or answer. A: market Q, each point one cartel and one announced quantity. API models sit on the diagonal (copy); Sonnet, Opus and Fable sit on the reaction curve (exact best reply); points at the bottom are firms that stay. B: all markets, partial claims, partner source.

1 of 8; Fig. 4B). Both firms choose at once, so this shows two regimes, recovery and price war, not that one choice causes the other.

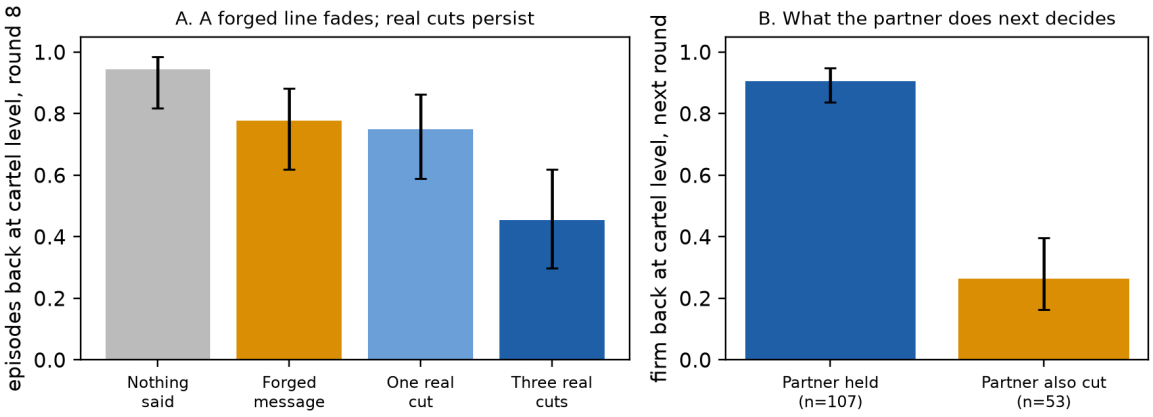


Figure 4: Persistence in full episodes. A: share of episodes back at the cartel level (price index at least 0.4) at round 8, cheap models. B: next-round recovery after a cut on a forged claim, by what the partner did. 95% Wilson intervals.

Three-firm studies by coauthors. Sumin Hong ran a three-firm quantity market with a local 9B model and two generated histories. A greedy offer with the offerer's output unchanged left the price at 52 and was followed in 0 of 5 cases in both histories. More output with no offer sent the price to cost (4) in both, with adherence in 4 of 5 cases; once the extra output ended the price rose only to 20. Chirag Chivate's reaction studies in two environments found that an equal offer from a trusted insider was followed in 5 of 5 cases and an unequal one in 0 of 5. Public replies voiced suspicion in 0 of 32 pairs in both environments; private answers flagged it in 15 of 32 (environment 1) and 6 of 32 (environment 2). Each environment rests on one generated history, so we report no rates from these data and use them as converging evidence only.

5 Discussion

Words start a response; actions decide how long it lasts. One sentence in the partner's name moves the firm at once, regardless of the claim's size, and fades by round 8 unless the partner's behavior backs it. A simple rule fits: an announcement is evidence about the partner's next move, and observation corrects it. We measure behavior, not belief, so this is an interpretation.

Capability changes how a claim is used. Weaker models imitate the announced number; Sonnet, Opus and Fable compute a reply. Opus and Fable also discount unverified outsiders in every run and Opus discounts an unverified label in both runs. Whether a stronger agent is harder or easier to steer therefore depends on the cue.

For defenders. (i) Authenticate the sender at the platform; a warning in the text protects only the strongest models. (ii) Watch actions as well as messages: in our three-firm studies, acceptance in words did not predict action and suspicion lived in private answers. (iii) A credible claim can break a harmful cartel and, equally, lawful coordination; because it fades, a forged claim is a trigger, not a remedy.

6 Limitations and dual use

Design. Single decisions on borrowed histories; the 17 cartels are reused across arms, so cells are not independent draws; temperature 0.3 gives near-duplicate episodes; firms are told to maximize profit. The subagent harness differs from the API (reasoning, batching) and subagent results vary between runs (Sonnet's outsider result, Fable's tag result); a first Fable sweep run did not see the label and was rerun, so only Opus's label result is repeatable.

Interpretation. Belief is inferred from behavior. The persistence link is a pattern across simultaneous choices. The three-firm data are one or two generated histories per environment, one 9B model and assistant-labeled responses.

What did not work. Best reply did not beat copying in API models (S1); response did not grow with claim size (S2); outsider and tag effects did not follow the predicted order (S3, S4); the label did not help with reasoning (B2, B3); a leak was not more convincing than an announcement (L3).

Dual use. A forged line can start a price war in lawful markets or break coordination among honest agents. The defense is authenticated messages and monitoring of actions. We release code and data so defenders can test their own agents.

References

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A Prediction record

We wrote 62 predictions before running: 24 held, 1 held in part, 33 failed, 4 not run. By study: ring joining 6 held, 1 failed; loyalty 1 held, 5 failed; remedies 3 held, 1 partly; forged records 2 held, 3 failed; source and action 0 held, 4 failed; wording, temperature, three-round cut 1 held, 2 failed; claim ladder and label 2 held, 3 failed; reasoning test 2 held, 1 failed; leak and proposal 3 held, 4 failed; controls and reasoning on four models 3 held, 3 failed; sweep 1 held, 4 failed. Every preregistration is dated before its run and in the repository.

B Contributions

[To be completed by the authors. Current assignment: A. Garodia, forged-claim, ring and remedy studies and cross-study analysis; S. Hong, three-firm talk versus action study; C. Chivate, three-firm reaction studies, controls and audits; A. Fu, repository and [role to confirm].]

C Earlier studies

Ring joining (six models, ten conditions, 16 wordings): GPT-4.1, Gemini Flash and Haiku 5.5 join a silently high-priced ring in 89 to 100% of rounds. Four explicit invitations lowered joining in GPT-4.1 (-46 points) and Haiku 5.5 (-31) and raised it in mini ($+34$). A verified warning from the start cut joining in GPT-4.1 from 54 to 9%. Loyalty appeals recruited nobody; a shutdown rule for reported rings raised reporting. Remedies in the three markets removed the stated agreement in 54 of 90 cases yet left prices near the cartel level in 21 of 32 informative cases. Full tables are in the repository.